

ε -Non-Deterministic Finite Automata

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NFA with ε -transitions

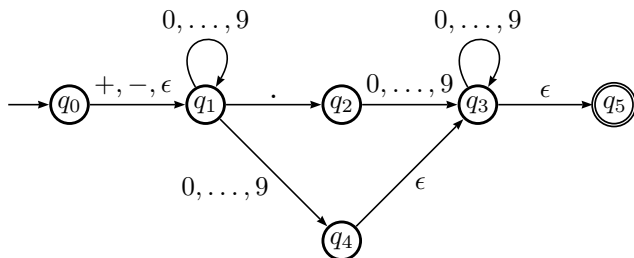
We shall now introduce another extension to NFA, called ε -NFA, which is a NFA whose labels can be the empty string, noted ε .

The interpretation of this new kind of transition, called ε -transition, is that the current state changes by following this transition *without reading any input*. This is sometimes referred as a **spontaneous transition**.

The rationale, that is, the intuition behind that, is that $\varepsilon a = a\varepsilon = a$, so recognising εa or $a\varepsilon$ is the same as recognising a . In other words, we do not need to read something more than a as input.

ϵ -NFA/Example

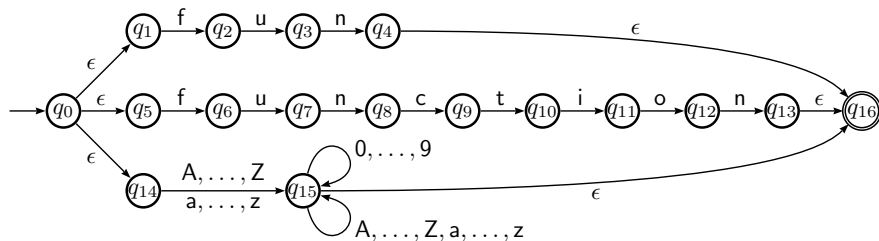
For example, we can specify signed natural and decimal numbers by means of the ϵ -NFA



This is not the simplest ϵ -NFA we can imagine for these numbers, but note the utility of the ϵ -transition between q_0 and q_1 .

ϵ -NFA

In the case of compilers or IDEs, ϵ -NFA enable the separate design of one NFA per token, then create an initial (respectively final) state connected to all their initial (respectively, final) states with an ϵ -transition. For instance, for keywords **fun** and **function** and identifiers:



ε -NFA

In order to perform a text search, once we have a single ε -NFA, we can

1. either remove all the ε -transitions and
 - 1.1 either create a NFA and then maybe a DFA;
 - 1.2 or create directly a DFA,
2. or use a formal definition of ε -NFA that directly leads to a recognition algorithm, just as we did for DFA and NFA.

Both methods assume that it is always possible to create an equivalent NFA, hence a DFA, from a given ε -NFA.

In other words, **DFA, NFA and ε -NFA have the same expressive power.**

ε -NFA

The first method constructs explicitly the NFA and maybe the DFA, while the second does not, at the possible cost of more computations at run-time.

Before entering into the details, we need to define formally an ε -NFA, as suggested by the second method.

The only difference between an NFA and an ε -NFA is that the transition function δ_E takes as second argument an element in $\Sigma \cup \{\varepsilon\}$, with $\varepsilon \notin \Sigma$, instead of Σ — but the alphabet still remains Σ .

ε -NFA/ ε -closure

We need now a function called **ε -close**, which takes an ε -NFA $\mathcal{E}$, a state q of $\mathcal{E}$ and returns all the states which are accessible in $\mathcal{E}$ from q with label ε (transitive closure of ε transitions).

The idea is to achieve a **depth-first traversal** of $\mathcal{E}$, starting from q and following only ε -transitions.

Let us call ε -DFS (“ ε -Depth-First-Search”) the function such that ε -DFS(q, Q) is the set of states reachable from q following ε -transitions and which is not included in Q , *Q being interpreted as the set of states already visited in the traversal*. The set Q ensures the termination of the algorithm even in presence of cycles in the automaton. Therefore, let

$$\varepsilon\text{-close}(q) = \varepsilon\text{-DFS}(q, \emptyset) \quad \text{if } q \in Q_E$$

where the ε -NFA is $\mathcal{E} = (Q_E, \Sigma, \delta_E, q_0, F_E)$.

ε -NFA/ ε -closure

Now we define ε -DFS as follows:

$$\varepsilon\text{-DFS}(q, Q) = \emptyset \quad \text{if } q \in Q \quad (1)$$

$$\varepsilon\text{-DFS}(q, Q) = \{q\} \cup \bigcup_{p \in \delta_E(q, \varepsilon)} \varepsilon\text{-DFS}(p, Q \cup \{q\}) \quad \text{if } q \notin Q \quad (2)$$

The ε -NFA page 3 leads to the following ε -closures:

$$\varepsilon\text{-close}(q_1) = \{q_1\}$$

$$\varepsilon\text{-close}(q_2) = \{q_2\}$$

$$\varepsilon\text{-close}(q_0) = \{q_0, q_1\}$$

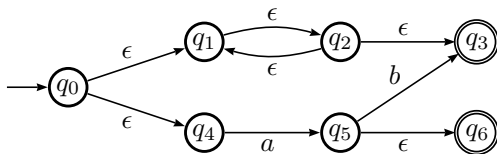
$$\varepsilon\text{-close}(q_3) = \{q_3, q_5\}$$

$$\varepsilon\text{-close}(q_5) = \{q_5\}$$

$$\varepsilon\text{-close}(q_4) = \{q_4, q_3, q_5\}$$

ε -NFA/ ε -closure

Consider, as a more difficult example, the following ε -NFA $\mathcal{E}$:



$$\begin{aligned}\varepsilon\text{-close}(q_0) &= \varepsilon\text{-DFS}(q_0, \emptyset) && \text{since } q_0 \in Q_E \\ &= \{q_0\} \cup \varepsilon\text{-DFS}(q_1, \{q_0\}) \cup \varepsilon\text{-DFS}(q_4, \{q_0\}) && \text{by eq. 2} \\ &= \{q_0\} \cup \left(\{q_1\} \cup \bigcup_{p \in \delta_E(q_1, \varepsilon)} \varepsilon\text{-DFS}(p, \{q_0, q_1\}) \right) && \text{by eq. 2} \\ &\quad \cup \left(\{q_4\} \cup \bigcup_{p \in \delta_E(q_4, \varepsilon)} \varepsilon\text{-DFS}(p, \{q_0, q_4\}) \right) && \text{by eq. 2}\end{aligned}$$

ε -NFA/ ε -closure

$$\begin{aligned}\varepsilon\text{-close}(q_0) &= \{q_0\} \cup \left(\{q_1\} \cup \bigcup_{p \in \{q_2\}} \varepsilon\text{-DFS}(p, \{q_0, q_1\}) \right) \\ &\quad \cup \left(\{q_4\} \cup \bigcup_{p \in \emptyset} \varepsilon\text{-DFS}(p, \{q_0, q_4\}) \right) \\ &= \{q_0\} \cup (\{q_1\} \cup \varepsilon\text{-DFS}(q_2, \{q_0, q_1\})) \cup (\{q_4\} \cup \emptyset) \\ &= \{q_0, q_1, q_4\} \cup \varepsilon\text{-DFS}(q_2, \{q_0, q_1\}) \\ &= \{q_0, q_1, q_4\} \cup \left(\{q_2\} \cup \bigcup_{p \in \delta_E(q_2, \varepsilon)} \varepsilon\text{-DFS}(p, \{q_0, q_1, q_2\}) \right)\end{aligned}$$

ε -NFA/ ε -closure

$$\begin{aligned}\varepsilon\text{-close}(q_0) &= \{q_0, q_1, q_4\} \cup \left(\{q_2\} \cup \bigcup_{p \in \{q_1, q_3\}} \varepsilon\text{-DFS}(p, \{q_0, q_1, q_2\}) \right) \\&= \{q_0, q_1, q_2, q_4\} \cup \varepsilon\text{-DFS}(q_1, \{q_0, q_1, q_2\}) \\&\quad \cup \varepsilon\text{-DFS}(q_3, \{q_0, q_1, q_2\}) \\&= \{q_0, q_1, q_2, q_4\} \cup \emptyset \quad \text{by eq. 1, since } q_1 \in \{q_0, q_1, q_2\} \\&\quad \cup \left(\{q_3\} \cup \bigcup_{p \in \delta_E(q_3, \varepsilon)} \varepsilon\text{-DFS}(p, \{q_0, q_1, q_2, q_3\}) \right) \quad \text{by eq. 2} \\&= \{q_0, q_1, q_2, q_3, q_4\} \cup \bigcup_{p \in \emptyset} \varepsilon\text{-DFS}(p, \{q_0, q_1, q_2, q_3\}) \\&= \{q_0, q_1, q_2, q_3, q_4\}\end{aligned}$$

ε -NFA/ ε -closure

It is useful to extend ε -close to sets of states, not just states.

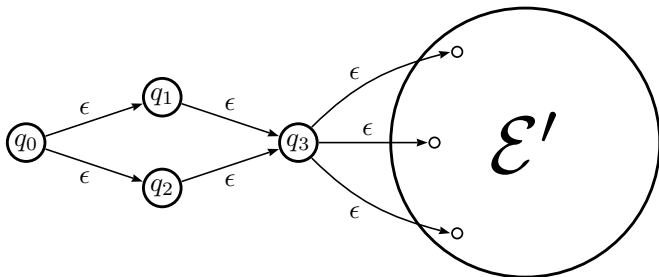
Let us note $\overline{\varepsilon\text{-close}}$ this extension, which we can easily define as

$$\overline{\varepsilon\text{-close}}(Q) = \bigcup_{q \in Q} \varepsilon\text{-close}(q)$$

for any subset $Q \subseteq Q_E$ where the ε -NFA is $\mathcal{E} = (Q_E, \Sigma, \delta_E, q_E, F_E)$.

ϵ -NFA/ ϵ -closure/Optimisation

Compute the ϵ -closure of q_0 in the following ϵ -NFA $\mathcal{E}$:



where the sub- ϵ -NFA $\mathcal{E}'$ contains only ϵ -transitions and all its Q' states are accessible from q_3 .

ε -NFA/ ε -closure/Optimisation

$$\begin{aligned}\varepsilon\text{-close}(q_0) &= \varepsilon\text{-DFS}(q_0, \emptyset) \\ &= \{q_0\} \cup \varepsilon\text{-DFS}(q_1, \{q_0\}) \cup \varepsilon\text{-DFS}(q_2, \{q_0\}) \\ &= \{q_0\} \cup (\{q_1\} \cup \varepsilon\text{-DFS}(q_3, \{q_0, q_1\})) \\ &\quad \cup (\{q_2\} \cup \varepsilon\text{-DFS}(q_3, \{q_0, q_2\})) \\ &= \{q_0, q_1, q_2\} \cup \varepsilon\text{-DFS}(q_3, \{q_0, q_1\}) \cup \varepsilon\text{-DFS}(q_3, \{q_0, q_2\}) \\ &= \{q_0, q_1, q_2, q_3, \} \cup (\{q_3\} \cup Q') \cup (\{q_3\} \cup Q') \\ &= \{q_0, q_1, q_2, q_3, \} \cup Q'\end{aligned}$$

We compute $\{q_3\} \cup Q'$ two times, that is, we traverse two times q_3 and all the states of $\mathcal{E}'$, which can be inefficient if Q' is big.

ε -NFA/ ε -closure/Optimisation

The way to avoid duplicating traversals is to change the definitions of ε -close and $\overline{\varepsilon}$ -close.

Dually, we need a new definition of ε -DFS and create function $\overline{\varepsilon}$ -DFS which is similar to ε -DFS except that it applies to set of states instead of one state:

$$\begin{aligned}\varepsilon\text{-close}(q) &= \varepsilon\text{-DFS}(q, \emptyset) && \text{if } q \in Q_E \\ \overline{\varepsilon}\text{-close}(Q) &= \overline{\varepsilon}\text{-DFS}(Q, \emptyset) && \text{if } Q \subseteq Q_E\end{aligned}$$

We interpret Q' in $\varepsilon\text{-DFS}(q, Q')$ and $\overline{\varepsilon}\text{-DFS}(Q, Q')$ as the set of states that have already been visited in the depth-first search.

Variables q and Q denote, respectively, a state and a set of states that have to be explored.

ε -NFA/ ε -closure/Optimisation

In the first definition we computed the *new reachable states*, but in the new one we compute the *currently reached states*. Then let us redefine ε -DFS this way:

$$\varepsilon\text{-DFS}(q, Q') = Q' \quad \text{if } q \in Q' \quad (1')$$

$$\varepsilon\text{-DFS}(q, Q') = \overline{\varepsilon\text{-DFS}(\delta_E(q, \varepsilon), Q' \cup \{q\})} \quad \text{if } q \notin Q' \quad (2')$$

Contrast with the first definition

$$\varepsilon\text{-DFS}(q, Q') = \emptyset \quad \text{if } q \in Q' \quad (1)$$

$$\varepsilon\text{-DFS}(q, Q') = \{q\} \cup \bigcup_{p \in \delta_E(q, \varepsilon)} \varepsilon\text{-DFS}(p, Q' \cup \{q\}) \quad \text{if } q \notin Q' \quad (2)$$

Hence, in (1) we return $\emptyset$ because there is no new state, i.e. none not already in Q' , whereas in (1') we return Q' itself.

ε -NFA/ ε -closure/Optimisation

The new definition of $\overline{\varepsilon\text{-DFS}}$ is not more difficult than the first one:

$$\overline{\varepsilon\text{-DFS}}(\emptyset, Q') = Q' \quad (3)$$

$$\overline{\varepsilon\text{-DFS}}(\{q\} \cup Q, Q') = \overline{\varepsilon\text{-DFS}}(Q, \varepsilon\text{-DFS}(q, Q')) \quad \text{if } q \notin Q \quad (4)$$

Notice that the definitions of $\varepsilon\text{-DFS}$ and $\overline{\varepsilon\text{-DFS}}$ are **mutually recursive**, i.e. they depend on each other.

In (2) we traverse states in *parallel* (consider the union operator), starting from each element in $\delta_E(q, \varepsilon)$, whereas in (2') and (4), we traverse them *sequentially* so we can use the information collected (currently reached states) in the previous searches.

ε -NFA/ ε -closure/Optimisation

Coming back to our example page 13, we find

$$\begin{aligned}\varepsilon\text{-close}(q_0) &= \varepsilon\text{-DFS}(q_0, \emptyset) && q_0 \in Q_E \\ &= \overline{\varepsilon\text{-DFS}}(\{q_1, q_2\}, \{q_0\}) && \text{by eq. (2')} \\ &= \overline{\varepsilon\text{-DFS}}(\{q_2\}, \varepsilon\text{-DFS}(q_1, \{q_0\})) && \text{by eq. (4)} \\ &= \overline{\varepsilon\text{-DFS}}(\{q_2\}, \overline{\varepsilon\text{-DFS}}(\{q_3\}, \{q_0, q_1\})) && \text{by eq. (2')} \\ &= \overline{\varepsilon\text{-DFS}}(\{q_2\}, \overline{\varepsilon\text{-DFS}}(\emptyset, \varepsilon\text{-DFS}(q_3, \{q_0, q_1\}))) && \text{by eq. (4)} \\ &= \overline{\varepsilon\text{-DFS}}(\{q_2\}, \varepsilon\text{-DFS}(q_3, \{q_0, q_1\})) && \text{by eq. (3)} \\ &= \overline{\varepsilon\text{-DFS}}(\{q_2\}, \{q_0, q_1, q_3\} \cup Q') \\ &= \overline{\varepsilon\text{-DFS}}(\emptyset, \varepsilon\text{-DFS}(q_2, \{q_0, q_1, q_3\} \cup Q')) && \text{by eq. (4)}\end{aligned}$$

ε -NFA/ ε -closure/Optimisation

$$\begin{aligned}\varepsilon\text{-close}(q_0) &= \varepsilon\text{-DFS}(q_2, \{q_0, q_1, q_3\} \cup Q') && \text{by eq. (3)} \\ &= \overline{\varepsilon\text{-DFS}}(\{q_3\}, \{q_0, q_1, q_2, q_3\} \cup Q') && \text{by eq. (2')} \\ &= \overline{\varepsilon\text{-DFS}}(\emptyset, \varepsilon\text{-DFS}(q_3, \{q_0, q_1, q_2, q_3\} \cup Q')) && \text{by eq. (4)} \\ &= \varepsilon\text{-DFS}(q_3, \{q_0, q_1, q_2, q_3\} \cup Q') && \text{by eq. (3)} \\ &= \{q_0, q_1, q_2, q_3\} \cup Q' && \text{by eq. (1')}\end{aligned}$$

The important thing here is that we did not compute (traverse) several times Q' . Note that some equations can be used in a different order and q can be chosen arbitrarily in equation (4), but the result is always the same.

Extended transition functions for ε -NFAs

The ε -closure allows to explain how a ε -NFA recognises or rejects a given input word.

In (2) we traverse states in *parallel* (consider the union operator), starting from each element in $\delta_E(q, \varepsilon)$, whereas in (2') and (4), we traverse them *sequentially* so we can use the information collected (currently reached states) in the previous searches.

Extended transition functions for ε -NFAs

The ε -closure allows to explain how a ε -NFA recognises or rejects a given input word.

Let $\mathcal{E} = (Q_E, \Sigma, \delta_E, q_0, F_E)$.

We want $\hat{\delta}_E(q, w)$ be the set of states reachable from q along a path whose labels, when concatenated, form the string w . The difference here with NFA's is that several ε can be present along this path, despite not contributing to w . For all state $q \in Q_E$, let

$$\hat{\delta}_E(q, \varepsilon) = \varepsilon\text{-close}(q)$$

$$\hat{\delta}_E(q, wa) = \overline{\varepsilon\text{-close}} \left(\bigcup_{p \in \hat{\delta}_E(q, w)} \delta_N(p, a) \right) \quad \text{for all } a \in \Sigma, w \in \Sigma^*$$

This definition is based on the regular identity $wa = ((w\varepsilon^*)a)\varepsilon^*$.

Extended transition functions for ε -NFAs/Example

Let us consider again the ε -NFA recognising natural and decimal numbers, at page 3, and compute the states reached on the input 5 . 6:

$$\hat{\delta}_E(q_0, \varepsilon) = \varepsilon\text{-close}(q_0) = \{q_0, q_1\}$$

$$\begin{aligned}\hat{\delta}_E(q_0, 5) &= \overline{\varepsilon\text{-close}}\left(\bigcup_{p \in \hat{\delta}_E(q_0, \varepsilon)} \delta_N(p, 5)\right) \\ &= \overline{\varepsilon\text{-close}}(\delta_N(q_0, 5) \cup \delta_N(q_1, 5)) = \overline{\varepsilon\text{-close}}(\emptyset \cup \{q_1, q_4\}) \\ &= \{q_1, q_3, q_4, q_5\}\end{aligned}$$

$$\begin{aligned}\hat{\delta}_E(q_0, 5.) &= \overline{\varepsilon\text{-close}}\left(\bigcup_{p \in \hat{\delta}_E(q_0, 5)} \delta_N(p, .)\right) \\ &= \overline{\varepsilon\text{-close}}(\delta_N(q_1, .) \cup \delta_N(q_3, .) \cup \delta_N(q_4, .) \cup \delta_N(q_5, .))\end{aligned}$$

Extended transition functions for ε -NFAs/Example

$$\hat{\delta}_E(q_0, 5.) = \overline{\varepsilon\text{-close}}(\{q_2\} \cup \emptyset \cup \emptyset \cup \emptyset) = \{q_2\}$$

$$\begin{aligned}\hat{\delta}_N(q_0, 5.6) &= \overline{\varepsilon\text{-close}}\left(\bigcup_{p \in \hat{\delta}_E(q_0, 5.)} \delta_N(p, 6)\right) \\ &= \overline{\varepsilon\text{-close}}(\delta_N(q_2, 6)) \\ &= \overline{\varepsilon\text{-close}}(\{q_3\}) \\ &= \{q_3, q_5\} \ni q_5\end{aligned}$$

Since q_5 is a final state, the string 5.6 is recognised as a number.

Subset construction for ε -NFAs

Let us present now how to construct a DFA from a ε -NFA such that both recognise the same language.

The method is a variation of the subset construction we presented for NFA: we must take into account the states reachable through ε -transitions, with the help of ε -closures.

Subset construction for ε -NFAs

Assume that $\mathcal{E} = (Q, \Sigma, \delta, q_0, F)$ is an ε -NFA. Let us define as follows the equivalent DFA $\mathcal{D} = (Q_D, \Sigma, \delta_D, q_D, F_D)$.

1. Q_D is the set of subsets of Q_E . More precisely, all accessible states of $\mathcal{D}$ are ε -closed subsets of Q_E , i.e. sets $Q \subseteq Q_E$ such that $Q = \varepsilon\text{-close}(Q)$.
2. $q_D = \varepsilon\text{-close}(q_0)$, i.e. we get the start state of $\mathcal{D}$ by ε -closing the set made of only the start state of $\mathcal{E}$.
3. F_D is those sets of states that contain at least one final state of $\mathcal{E}$, that is to say $F_D = \{Q \mid Q \in Q_D \text{ and } Q \cap F_E \neq \emptyset\}$.
4. For all $a \in \Sigma$ and $Q \in Q_D$, let $\delta_D(Q, a) = \overline{\varepsilon\text{-close}\left(\bigcup_{q \in Q} \delta_E(q, a)\right)}$

Subset construction for ε -NFAs/Example

Let us consider again the ε -NFA page 3. Its transition table is

ε	+	-	0, ..., 9	.	ε
$\rightarrow q_0$	$\{q_1\}$	$\{q_1\}$	$\emptyset$	$\emptyset$	$\{q_1\}$
q_1	$\emptyset$	$\emptyset$	$\{q_1, q_4\}$	$\{q_2\}$	$\emptyset$
q_2	$\emptyset$	$\emptyset$	$\{q_3\}$	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\emptyset$	$\{q_3\}$	$\emptyset$	$\{q_5\}$
q_4	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\{q_3\}$
$\#q_5$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

Subset construction for ε -NFAs/Example

By applying the subset construction to this ε -NFA, we get the table

$\mathcal{D}$	$+$	$-$	$0, \dots, 9$	$.$
$\rightarrow\{q_0, q_1\}$	$\{q_1\}$	$\{q_1\}$	$\{q_1, q_3, q_4, q_5\}$	$\{q_2\}$
$\{q_1\}$	$\emptyset$	$\emptyset$	$\{q_1, q_3, q_4, q_5\}$	$\{q_2\}$
$\#\{q_1, q_3, q_4, q_5\}$	$\emptyset$	$\emptyset$	$\{q_1, q_3, q_4, q_5\}$	$\{q_2\}$
$\{q_2\}$	$\emptyset$	$\emptyset$	$\{q_3, q_5\}$	$\emptyset$
$\#\{q_3, q_5\}$	$\emptyset$	$\emptyset$	$\{q_3, q_5\}$	$\emptyset$

Subset construction for ε -NFAs/Example

Let us rename the states of $\mathcal{D}$ and get rid of the empty sets:

$\mathcal{D}$	$+$	$-$	$0, \dots, 9$	$.$
$\rightarrow A$	B	B	C	D
B			C	D
$\#C$			C	D
D			E	
$\#E$			E	

Subset construction for ε -NFAs/Example

The transition diagram of $\mathcal{D}$ is therefore

